Solar ESS Co. Ltd.
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Which single-phase hybrid inverter is best?

10–16 kW single-phase hybrid inverters for Thai homes, compared on power, battery choice, protection, warranty and service

Solis · GoodWe · Deye · Sungrow · LuxPower · Growatt

1 The short answer

BEST OVERALL FOR HOMES

Solis S6-EH1P 12–16K

IP66 · < 4 ms · 2× surge

- Starts air cons and pumps easily
- Works with any approved 48 V battery
- Toughest casing, fastest backup
- 10-year warranty, good support

BEST FOR UP TO 12 KW

GoodWe ES 8–12K

IP66 · < 4 ms · 10 yrs

- 2× surge for 10 s for motors
- 48 V batteries, generator input
- Reliable brand, 10-year warranty

MOST ENERGY-MANAGEMENT FEATURES

Deye SUN-12–16K

100 A pass-through · 16 in parallel

- Very strong specs and energy management
- Generator input built in
- Claims can be slow in Thailand

2 Overall scorecard

Inverter	Power & surge	Battery choice	Protection & features	Warranty & service	Overall
Solis S6-EH1P 16K	★★★★★	★★★★½	★★★★★	★★★★½	4.8
GoodWe GW12K-ES-C10	★★★★☆	★★★★½	★★★★★	★★★★½	4.5
Deye SUN-16K-SG01LP1	★★★★★	★★★★★	★★★★☆	★★½☆☆	4.1
Sungrow SH10RS	★★★½☆	★★½☆☆	★★★★½	★★★★☆	3.6
LuxPower SNA12000	★★★★☆	★★★★☆	★★★☆☆	★★½☆☆	3.4
Growatt SPE 12000 ES	★★★★☆	★★★★☆	★★☆☆☆	★★★☆☆	3.2

Solar ESS assessment, 5 = best, for single-phase homes of 10–16 kW. Based on makers' datasheets and our field experience in Thailand. Sol-Ark and EG4 are left out: they are built for the US 120/240 V split-phase supply and are not offered in Thailand.

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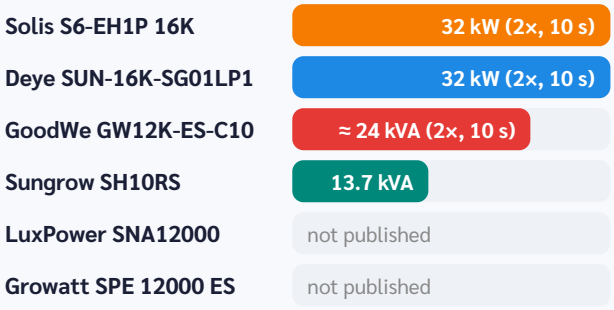
3 Power and surge

🔌 Rated single-phase output



Largest single-phase model in each range.

❄️ Short surge for starting air cons and pumps



A high surge rating lets compressors start without tripping the inverter.

4 Technical specifications

Item	Solis S6-EH1P 16K	GoodWe GW12K-ES-C10	Deye SUN-16K-SG01LP1	Sungrow SH10RS	LuxPower SNA12000	Growatt SPE 12000 ES
Single-phase sizes	12/14/16 kW	8/10/12 kW	12/14/16 kW	8/10 kW	12 kW	12 kW
Battery type	48 V LV	48 V LV	48 V LV	HV (Sungrow)	48 V LV	48 V LV
Max solar panels	22.4 kWp	≈ 2× (10K: 20 kWp)	25.6 kWp	20 kWp	24 kWp	15 kWp
MPPT	3 × 40 A	2 × 32 A	3 × 32 A	4	2 × 35 A	2 × 27 A
Backup switch	< 4 ms	< 4 ms	UPS-class	< 10 ms	< 10 ms	n/p
Max efficiency	97.6%	97.6%	97.6%	97.5%	n/p	96.5%
Enclosure	IP66	IP65–66	IP65	IP65	n/p	IP20 (indoor)
Parallel units	10	6–16	16	—	3	9
Generator input	✔	✔	✔	—	✔ 15 kW	✔
Warranty	10 yrs	10 yrs	5 or 10 yrs	10 yrs (5+5)	n/c †	n/p

n/p = not published · n/c = not confirmed. † Warranty for LuxPower varies by dealer; one listing shows 2 years. GoodWe parallel count differs between listings (6–16). Solis 12–16K figures from the maker's preliminary datasheet. Confirm Thai availability and PEA certificate before ordering.

5 Warranty and after-sales service

Inverter	Service	What to expect in Thailand
Solis S6-EH1P 16K	★★★★½	Responsive back-office support through official distribution; strong off-grid and unbalanced-load support; practical remote app.
Deye SUN-16K-SG01LP1	★★½☆☆	Rich energy-management features, but claims are handled through regional distributors and can take a long time.
Sungrow SH10RS	★★★★☆	Large global brand with structured warranty; best with Sungrow's own high-voltage batteries.
GoodWe GW12K-ES-C10	★★★★½	Established brand with a 10-year standard warranty, structured claims and a good monitoring app.
LuxPower SNA12000	★★½☆☆	Widely used for off-grid; service via local dealers, warranty terms vary by dealer.
Growatt SPE 12000 ES	★★★☆☆	Off-grid unit for indoor installation; service via local dealers.

6 Three golden rules before buying

🔋 Match the battery voltage
Low-voltage (48 V) inverters need 48 V batteries; high-voltage inverters need high-voltage batteries. Never mix them.

🔗 Closed-loop communication
Check the battery is on the inverter maker's approved list and talks to it over CAN or RS485.

❄️ Check surge, not just kW
Air cons and deep-well pumps need high starting power. Check the surge rating and pass-through current.

7 Our recommendation

Most single-phase homes, 12–16 kW → **Solis S6-EH1P 12–16K** with 48 V LiFePO4 batteries.

Advanced energy management, large parallel systems → **Deye SUN-12–16K**, with a plan for slower warranty claims.

Homes up to 12 kW → **GoodWe ES 8–12K**. Premium 10 kW with high-voltage battery and AFCI: Sungrow SH10RS.

Simple off-grid, indoor install → **LuxPower SNA12000** or **Growatt SPE 12000 ES**, installed indoors, with a clear written warranty from the dealer.

Thai rules: home rooftop systems up to 10 kWp qualify for the PEA-connected tax deduction. Very large single-phase loads may need a bigger meter or a 3-phase supply. Check with PEA before you buy.

Sources: solisinverters.com (S6-EH1P 12–16K-L), deyeinverter.com (SUN-12/14/16K-SG01LP1, 2026), GoodWe ES-C10 (GW10K/GW12K), Sungrow SH10.0RS / SH8.0RS, LuxPower SNA12000, Growatt SPE 12000 ES, PEA (pea.co.th). October 2026. Confirm against the current datasheet before ordering.